Mobile Application Development – CA3: Review 1

Idea & AI Use-Case Proposal

Milestone: Problem definition, AI/ML use case, and project scope

Section	Details
Group Number	20
Team Members	Ruksar Khatun (USN: 25MCAR0115) Sahil Dcunha (USN: 25MCAR0117) Yashas B Reddy (USN: 25MCAR0231)
Title	TripWise: AI-Powered Smart Travel Planner for India
App Name	TripWise (AI assistant: TripMate AI)
Proposed AI/ML Feature	**Recommendation + intelligent decision support:** TripMate AI (Google Gemini) suggests destinations, activities, stays and restaurants, answers travel questions and generates a personalised day-by-day itinerary from destination, days, budget, pace and interests.
Why AI is Needed	Trip planning mixes many factors (budget, days, interests, season, group size) spread across separate apps for places, routes, hotels and food. Fixed filters cannot write personalised plans or answer open questions; an AI model grounded in our curated data can, in seconds.
Target Users	Students, young travellers, families and first-time visitors planning trips within India; budget-conscious Android users who want quick plans and costs in ₹.
Add-ons as per Remarks	India-only scope (16 destinations, prices in ₹); assistant named TripMate AI; Google Maps hand-off instead of a paid Maps API.

Remarks:

Ruksar Khatun — 25MCAR0115

Sahil Dcunha — 25MCAR0117

Yashas B Reddy — 25MCAR0231

Signature of Members

26/11/26

Signature of Faculty In charge

Mr. Aldrin Jose Antony

Mobile Application Development – CA3: Review 2

AI/ML Model & Data Review

Milestone: Dataset preparation, model selection, and training approach

Section	Details
Group Number	20
Team Members	Ruksar Khatun (USN: 25MCAR0115) Sahil Dcunha (USN: 25MCAR0117) Yashas B Reddy (USN: 25MCAR0231)
Title	TripWise: AI-Powered Smart Travel Planner for India
App Name	TripWise (AI assistant: TripMate AI)
Dataset Used	**Curated catalog** made by the team from public travel sources: 16 Indian destinations, 80 attractions, local food, best time, typical costs in ₹ and licensed photos (Wikimedia Commons, Unsplash). The user's own trip and preferences are added per request.
Data Preprocessing	Catalog checked against a strict schema; costs normalised to ₹; no invented ratings; image licences verified. Prompts include the relevant destination and attractions, the last 20 chat turns, the trip plan and preferences.
Model / Algorithm Chosen	**Pre-trained LLM via API:** Google Gemini 3.5 Flash, with automatic fallback to other Flash models. Chosen because no labelled travel dataset exists to train our own model, and it supports streaming chat and structured JSON output.
Training & Validation Approach	No custom training or train/test split (pre-trained model). Every structured AI answer is schema-validated and normalised; invalid output is retried once, then replaced by a curated plan. 43 automated backend tests plus live API checks.
Model Performance (so far)	Accuracy/precision/recall not applicable (generative model, no labelled test set). Live tests: valid streamed answers, and an AI itinerary that respected a vegetarian preference and used our attraction IDs.
Integration Plan	**Cloud API via our backend:** Android app → HTTPS → Node.js/Express → Gemini. API key stays on the server; daily limit of 100 AI requests per user.
Add-ons as per Remarks	Added model fallback after gemini-2.5-flash was retired; stay/restaurant cards; offline curated itinerary fallback.

Evaluation – AI Integration & Performance (partial)

Remarks:

Ruksar Khatun — 25MCAR0115
Sahil Dcunha — 25MCAR0117
Yashas B Reddy — 25MCAR0231

Signature of Members

06/4/26

Signature of Faculty In charge
Mr. Aldrin Jose Antony

Mobile Application Development – CA3: Review 3

App Development & Functionality Review

Milestone: Core app build and functional integration of the AI feature

Section	Details
Group Number	20
Team Members	Ruksar Khatun (USN: 25MCAR0115) Sahil Dcunha (USN: 25MCAR0117) Yashas B Reddy (USN: 25MCAR0231)
Title	TripWise: AI-Powered Smart Travel Planner for India
App Name	TripWise (AI assistant: TripMate AI)
Features Implemented	Sign-in/sign-up, guest mode, Home, Explore, Destination details, Things to do, Google Maps shortcuts, TripMate AI chat and history, AI itinerary planner, My Trips with itinerary editing, budget and expenses, Saved, Profile.
Tech Stack Used	**Android:** Kotlin, Jetpack Compose, Material 3, Hilt, Room, Retrofit. **Backend:** Node.js, Express, TypeScript, PostgreSQL. **AI:** Google Gemini API. **CI:** GitHub Actions.
AI Feature Functionality	**Input:** question, quick action or planner form, plus destination/trip, budget, days and interests. **Processing:** backend builds a grounded prompt and calls Gemini, then validates the result. **Output:** streamed reply, day-by-day plan, or stay/restaurant cards with Google Maps links.
Error Handling / Edge Cases	Form validation; clear messages for no connection, timeout, quota or AI unavailable, with Retry and Stop; invalid AI output falls back to a labelled "Basic plan"; offline browsing and planning.
Known Issues / Bugs	Backend not hosted online yet, so accounts and TripMate AI need it running locally; APK is a debug build; chat history is not synced.
Add-ons as per Remarks	Guest mode for phone use without a server; catalog bundled in the app; clearer errors and faster timeouts.

Evaluation – Application Functionality

Remarks:

Ruksar Khatun — 25MCAR0115
Sahil Dcunha — 25MCAR0117
Yashas B Reddy — 25MCAR0231

Signature of Members

06/11/26

Signature of Faculty In charge
Mr. Aldrin Jose Antony

Mobile Application Development – CA3: Review 4

UI/UX & AI Output Presentation Review

Milestone: Interface design and clarity of AI-generated results

Section	Details
Group Number	*20*
Team Members	Ruksar Khatun (USN: 25MCAR0115) Sahil Dcunha (USN: 25MCAR0117) Yashas B Reddy (USN: 25MCAR0231)
Title	TripWise: AI-Powered Smart Travel Planner for India
App Name	TripWise (AI assistant: TripMate AI)
Screens Designed	Sign in, Sign up, Home, Explore, Destination details, Things to do, TripMate AI chat, Trip preferences, Chat history, Planner, My Trips, Trip details (Overview/Itinerary/Budget), Saved, Profile.
Navigation Flow	Bottom bar: Home, Explore, Saved, My Trips, Profile. TripMate AI opens from Home, Destination details ("Where to stay / eat"), a trip, and the planner; "Plan with AI" opens the planner.
AI Output Presentation	Streamed formatted chat replies; recommendation cards with area, price in ₹, reason and "Open in Google Maps"; itinerary by day with times and costs, and a label showing AI or curated plan.
Interactive Components Used	Buttons, chips, text fields, segmented buttons, steppers, bottom sheet, dialogs, tabs, cards, lists, carousels, loading indicators, snackbars.
Responsiveness Check	Pixel 7 emulator (1080 × 2400, Android 14), light and dark themes; tablet layout (navigation rail, list–detail) implemented.
Add-ons as per Remarks	Brand colours and TripWise logo applied; labels shortened to fit phones; map replaced by a Google Maps card.

Evaluation – UI/UX Design

Remarks:

Ruksar Khatun — 25MCAR0115

Sahil Dcunha — 25MCAR0117

Yashas B Reddy — 25MCAR0231

Signature of Members

06/11/26

Signature of Faculty In charge

Mr. Aldrin Jose Antony

Mobile Application Development – CA3: Review 5

Final Deployment & Demo Review

Milestone: Deployed application, final AI performance, and live demo

Section	Details
Group Number	20
Team Members	Ruksar Khatun (USN: 25MCAR0115) Sahil Dcunha (USN: 25MCAR0117) Yashas B Reddy (USN: 25MCAR0231)
Title	TripWise: AI-Powered Smart Travel Planner for India
App Name	TripWise (AI assistant: TripMate AI)
Deployment Details	**APK:** https://github.com/ruksar-khatun/TripWise-v2/tree/ci-apk. **Code:** https://github.com/ruksar-khatun/TripWise-v2. Backend ready for Google Cloud Run (Dockerfile); currently run locally.
Final AI Model Performance	No accuracy metric (generative model). Since Review 2: model fallback, stay/restaurant cards, offline fallback plans; all automated tests pass.
Final Feature Summary	Accounts and guest mode, 16 destinations, Google Maps hand-off, TripMate AI with preferences and history, AI itinerary, trips, budget, saved places, profile.
Demo Notes	Guest mode → Explore → destination → generate and save a plan → budget; then sign in and use TripMate AI; promo video.
Project Report Submitted	Yes: README documentation at https://github.com/ruksar-khatun/TripWise-v2#readme
Add-ons as per Remarks	Downloadable APK and project ZIP; promo video; fixes for running on a phone without a server.

Final Evaluation – CA3 Rubric (Total 15 Marks)

Ruksar Khatun — 25MCAR0115
Sahil Dcunha — 25MCAR0117
Yashas B Reddy — 25MCAR0231

Signature of Members

06/11/26

Signature of Faculty In charge
Mr. Aldrin Jose Antony